

Technology Design for Epistemic Niche Construction in the AI Era*

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ABSTRACT

Generative AI is restructuring the epistemic environments in which humans think, learn, and produce knowledge. This paper argues that *epistemic niche construction*—the process by which human agents build and modify the representations, tools, practices, and values that sustain their knowledge work, and are cognitively reshaped by what they build—provides a productive framework for understanding this restructuring and for guiding technology design in education. To motivate the analysis, I identify three features that make generative AI qualitatively different from prior cognitive technologies as a modifier of epistemic niches: representational opacity, generativity, and substrate asymmetry. Each produces distinctive feedback dynamics that can either enrich or erode human epistemic niches. Drawing on emerging literature across contexts, I present different enrichment and erosion mechanisms and discuss implications for designing epistemic environments for learning and knowledge work in the AI era.

1 Introduction

In January 2026, CNN profiled Alpha School, a private school network that asks: “What if I told you this school had no teachers?”¹ This school’s approach is to use AI to “speed-teach core academic subjects in just two hours a day,” and devote the rest of the school day to “workshops in financial literacy, communication, and problem solving.”

Alpha School is one of many emerging models of integrating AI in education. It draws on persistent anxieties in American education—declining test scores, unfavorable international comparisons, the recurring sense that public schools are not doing enough (Berliner & Biddle, 1996)—anxieties that have long fueled demand for alternatives to public education and channeled reform energy toward standards, accountability, and efficiency (Mehta, 2013). The specific diagnosis these models act on is familiar: that education’s primary job is content delivery, and AI’s contribution is to do it faster. The two-hour AI curriculum operates on this logic. What is school for if it is not to help students absorb as much information as possible? Despite decades of efforts to reorganize learning around inquiry, design, and knowledge creation—from constructionism (Holbert et al., 2020) to maker education (Halverson & Sheridan, 2014) to knowledge building (Scardamalia & Bereiter, 2021)—these approaches remain peripheral to content-focused curriculum reforms.

Throughout the history of educational technology, new technologies have been absorbed into existing institutional structures, leaving the underlying logic of education intact (Cuban, 1990; Reich, 2020). Technology itself will do little to disrupt the existing logic or system inertia. Without examining the logic, technological disruption will at best under-deliver the promise and in worse situations cause harms to individuals and the fabric of the system.

Designing technology for learning has never been primarily an engineering problem. It is a conceptual, pedagogical, political, and philosophical problem about what kind of learning we want to enable and what it means to learn (Philip & Sengupta, 2021). The history of educational technology is a history of choices about what cognitive work to support, what to cognitively offload, and what to leave to the learner. Each choice reshapes the environment in which learning happens. Educational technologies embed assumptions, intentionally or not, about knowledge, pedagogy, and human capacity that deserve critical scrutiny (Selwyn, 2013). A technology that automates essay grading makes a different bet about the nature of writing than a technology that helps students revise their own arguments. A technology that delivers personalized content at an “optimal” pace operates on a different model of learning than one that helps a community build shared knowledge. Both are functional. They give rise to different epistemic environments for learning.

Capabilities of generative AI makes this examination of the underlying logic of education urgent, as they are reshaping informational environments with profound consequences for learning. LLM’s token-by-token inference is now cheap and ubiquitous, from chatbots to AI overviews, making AI-generated content pervasive for many cognitive tasks. Agentic AI

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¹<https://www.cnn.com/2026/01/29/politics/alpha-school-trump-ai-teaching>

systems act on behalf of users, able to finish long-range tasks without user input. For a growing variety of tasks, the human role shifts from composing to evaluating (Shen & Tamkin, 2026). AI's epistemic dispositions, embedded in training data and optimization objectives, shape outputs in ways that go undetected and unchallenged (Bender et al., 2021; B. Chen et al., 2025). When millions of knowledge workers depend on the same few foundation models, the range of cognitive strategies and conceptual repertoires across a population narrows even as individual output quality improves (Doshi & Hauser, 2023; Messeri & Crockett, 2024). Human thinking is being shaped by AI-infused environments at a scale and speed that outpaces awareness.

In education, the dominant responses to this evolving landscape remain at the surface level: detect AI-generated text, restrict access, add AI use policies, replace written essays with oral exams. These responses address symptoms without addressing the structural change that produces them. What is at stake is AI fundamentally reshapes the environment in which humans form beliefs and build knowledge. To respond at the level at which AI actually operates, we need to understand how AI restructures the epistemic environments in which humans think, learn, and produce knowledge.

This paper applies *epistemic niche construction* as an analytical framework to address the gap. The concept originates in biological niche construction theory (Odling-Smee et al., 2003), which holds that organisms modify their environments in ways that feed back to reshape the organisms themselves. Sterelny (2003, 2010, 2012) focused this framework on its epistemic dimension, centering a specific claim: humans actively construct the representations, practices, and values that sustain their knowledge work, and are cognitively reshaped by what they construct. An *epistemic niche* is the set of these conditions—the shared notations, inquiry routines, and epistemic standards within which a community's members pursue epistemic goals and develop their cognitive capacities. Epistemic niches operate at every scale, from the practices of individual researchers and professional communities, to the representational systems shared across societies and cultures, to the cumulative cognitive inheritance that distinguishes *Homo sapiens* as a species.² The concept foregrounds humans as constructors of their epistemic environments, and it identifies the feedback loop through which the constructed niche reshapes the constructor's epistemic endeavors. Situating AI's impact within this framework reframes the design question. The question is no longer *How does technology support learning?* but *How does technology support humans in constructing epistemic niches that sustain their capacity to learn, inquire, and produce knowledge?*

The paper proceeds as follows. Section 2 introduces the epistemic niche construction framework, grounding it in biological niche construction theory and specifying the representations, practices, and values that constitute an epistemic niche. Section 3 uses three features of generative AI to illustrate what makes it qualitatively different from prior cognitive technologies as a modifier of epistemic niches: representational opacity, generativity, and substrate asymmetry. Section 4 traces two possibilities, enrichment and erosion, that these features produce depending on how the niche is designed, with design implications for each. Section 5 discusses broader implications for educational change, technology design, and empirical research, and proposes a research agenda centered on cognitive ethnography of AI-integrated epistemic niches. The core argument is that humans are, in Sterelny's (2003) phrase, "epic epistemic niche constructors," and that the central challenge of AI in education is to preserve and strengthen this constructive capacity rather than destroy it.

2 Epistemic Niche Construction

2.1 Niche construction in biology

Niche construction as a concept originates in evolutionary biology as a corrective to a one-directional picture of adaptation. In the classical formulation of an ecological niche associated with Hutchinson (1957), a *niche* is the *n*-dimensional space of environmental conditions within which a population can persist—a space the organism occupies. Lewontin (1983) added the missing direction: organisms also shape their environments, constructing the conditions in which they and their descendants live. Odling-Smee and colleagues (2003) developed the idea under the term *niche construction*. Organisms modify their niches, and those modifications change the selection pressures acting on themselves and their descendants. The beaver builds a dam, creating a pond that selects for aquatic adaptations in subsequent generations. Earthworms alter soil chemistry, changing the selective environment for future earthworms. Termites build mounds that regulate temperature and humidity, structuring the developmental conditions for future colonies.

What distinguishes niche construction from ordinary environmental modification is that modifications persist and feed back to alter selection pressures on the modifying organisms and their descendants. Odling-Smee et al. (2003) argued that this feedback makes niche construction a second evolutionary process operating alongside natural selection. The modifications can also be inherited. Through ecological inheritance (Laland et al., 2016; Odling-Smee et al., 2003), offspring inherit not just genes but also the modified environment their parents constructed. This second inheritance channel, ecological alongside genetic, means that the developmental conditions for the next generation are partly determined by the constructive activities of the current generation.

²An individual's epistemic innovation gains its force through broader adoption. A novel practice is useful when more people adopt it and build normative standards and infrastructures around it.

2.2 Extension to cognitive and epistemic niche construction

The concept of niche construction has been extended to humans, with special attention to human cultures (Odling-Smee et al., 2003). Cultural practices modify environments far faster than genetic evolution, and cultural transmission allows modifications to accumulate across generations independently of genetic inheritance. Several scholars have advanced the cognitive and epistemic dimensions of human niche construction.

Language functions as cognitive niche construction (Clark, 2006). The claim goes beyond the familiar idea that language shapes thought. Language creates public, persisting representational structures—written texts, formal notations, diagrams—that become part of the material environment in which cognition operates (Donald, 1993). A community that develops written argument acquires cognitive capacities unavailable to one that has not, and those capacities enable further representational innovation. The feedback loop between representational environment and cognitive capacity is the same one identified in biological niche construction, now operating at the level of thought. Adults structure learning environments—organized play, apprenticeship, formal instruction—that shape young people’s cognitive development along culturally specific trajectories. Flynn (2013) calls this developmental niche construction. Each generation inherits a structured learning environment, develops within it, and modifies it for the next. The mechanism matters here because it is how epistemic niches are transmitted across generations.

Building on these threads, Sterelny (2003, 2010, 2012) focused the framework on a specific dimension: the epistemic. “Much niche construction is epistemic,” Sterelny writes, “making cognitive tools and assembling other informational resources that support and scaffold intelligent action” (2010, p. 465). An *epistemic niche* is the set of representational, practical, and normative conditions that a community constructs to support its knowledge work, and within which its members’ cognitive capacities develop.³ The niche is partly inherited and partly constructed, and the constructing feeds back to reshape the agents occupying it. Humans are, in Sterelny’s (2003) phrase, “epic epistemic niche constructors.”⁴ The capacity to progressively restructure our own informational environments across generations is one of the defining features of our species.

Mathematical notation illustrates the feedback loop between representational environment and cognitive capacity. Arabic numerals make certain operations tractable that Roman numerals make nearly impossible (Norman, 2014). The notation reshapes what its users can think by changing the representational environment in which their brains operate. Positional notation gave its adopters access to cognitive operations that were previously unavailable, which opened the way to further mathematical development. The representational innovation fed back to change what anyone inhabiting the modified environment could do.

Once adopted, representational innovations also accumulate institutional scaffolding that resists displacement. The decimal system persists because an entire infrastructure of practices, tools, and training has been built around it, even though base-10 has no mathematical advantage over alternatives (Patel, 2026). Each generation inherits not just the symbols but the ecology of competences organized around them, an instance of ecological inheritance operating at the level of representational tools.

Epistemic niche construction operates through two mechanisms that are central for what follows (Sterelny, 2012). First, epistemic niches enable *downstream epistemic engineering*, where modifications to the epistemic environment shape the cognitive development of those who enter it. Parents organize play, teachers structure curricula, lab directors establish inquiry routines, organizations codify knowledge practices—all of which determine what cognitive challenges newcomers encounter, in what order, and with what supports. Second, epistemic niches support *cumulative construction*, where each round of modification builds on the last, creating a ratchet effect in which capacities accumulate beyond what any individual could achieve alone. Writing enables mathematics that enables physics that enables engineering; a research group’s analytic conventions build semester by semester into a shared intellectual infrastructure. These two mechanisms highlight that human cognitive capacities depend on structured environmental resources that communities construct, inherit, and transmit.

One feature of epistemic niches that matters for what follows is their timescale. Unlike biological niches, which shift across generations, epistemic niches are volatile. The inquiry practices of a learning community can be established in weeks and can collapse just as quickly if the conditions sustaining them are removed. A classroom that has built up sophisticated practices of evidence-based argumentation can lose those practices within a semester if AI tools make the productive effort unnecessary. This volatility makes epistemic niches both more powerful and more fragile than their biological counterparts.

This fragility makes the question of how AI modifies epistemic niches urgent. The question extends beyond whether AI helps learners perform better within a given environment. It is what AI does to their capacity to construct and maintain epistemic environments in the first place. A technology that takes over the construction work, generating representations, performing

³For epistemic niche construction to do explanatory work, the components of the niche also need to be specified. One approach draws on Thomas Kuhn’s disciplinary matrix, where a community’s epistemic niche consists of shared symbolic generalizations, methodological assumptions, epistemic values, and exemplars that organize collective inquiry (De Benedetto & Luchetti, 2025). These components may be applicable beyond scientific communities. The present paper specifies components at a different grain, focusing on representational infrastructure, inquiry practices, and epistemic norms as the elements that AI modifies.

⁴What distinguishes humans is the cumulative, culturally transmitted character of their epistemic modifications (Sterelny, 2003). Epistemic niche construction is not unique to humans. Chimpanzees modify their environments with tools that extend what they can perceive and manipulate. What sets humans apart is that these modifications accumulate across generations through cultural transmission rather than being reinvented by each individual.

inquiry routines, applying evaluative standards, may serve immediate performance goals while undermining the constructive capacity on which sustained knowledge work depends. A technology that supports construction, helping communities build better representations, strengthen their inquiry practices, and make their epistemic values visible and negotiable, serves a different function. Which of these roles AI plays depends on how the epistemic niche is redesigned.

3 Epistemic Niche Construction in the AI Era

Every major cognitive technology in human history has restructured the epistemic niches in which humans think. Writing, mathematical notation, the printing press, digital media—each represented a qualitative rupture, introducing new representational media that modified the epistemic environment and reshaped human cognition (Donald, 1993).

Yet across these transitions, certain features of the niche held steady. The products of knowledge work, from ancient manuscripts to modern codebases, remained inspectable; a competent practitioner could trace how a claim was built and evaluate whether it holds. And no matter how powerful the tool, human agents remained central to the generation of epistemic content. Tools amplified what humans could do, but the generative epistemic work—formulating questions, constructing arguments, building explanations—stayed with them.

Generative AI puts each of these under pressure. Three features of generative AI create this pressure. Representational opacity removes the accessibility on which knowledge validation depends. Generativity shifts the locus of epistemic production from human agents to AI. And substrate asymmetry—the material differences between how brains and computers process information—means that AI performs epistemic work under conditions fundamentally unlike those of human agents. These three are not exhaustive, but they mark where generative AI breaks from every cognitive technology that came before.

3.1 Representational opacity

Prior cognitive technologies stored knowledge in human-readable symbolic form: text, notation, diagrams, databases. A competent reader can trace how a claim in any of these formats was constructed and evaluate whether it holds. Large language models store information in a fundamentally different format—weights distributed across billions (or even trillions) of parameters. No individual weight corresponds to an identifiable concept. The system’s knowledge exists only as patterns across the entire weight space. The continuous, distributed nature of these representations lacks the discrete compositionality that makes symbolic representations interpretable (Ferrone & Zanzotto, 2020). No human, however expert, could trace how a given output was produced.

But opacity itself is not new in epistemic practice. A physicist using a particle collider may not understand the engineering of every detector subsystem. A biologist relying on mass spectrometry may treat the instrument as a black box. Statistical methods have long produced results that many of their users cannot fully explain. The relevant distinction is between opacity that is *practical*—where a process could in principle be followed by a sufficiently informed human but is not followed in practice for practical reasons—and what Humphreys (2009) calls *essential epistemic opacity*, where the process cannot in principle be followed, regardless of the human’s expertise or available time. The collider is practically opaque; an engineer with enough time can trace its causal chain. A statistical regression is practically opaque; its logic is fully formalizable and inspectable. An LLM is *essentially* opaque—billions of parameters interact through nonlinear transformations that no human could trace even with unlimited time, because the representations are continuous and distributed rather than discrete and compositional. This places AI in a novel epistemic category, distinct from both scientific instruments (whose reliability rests on understood causal mechanisms) and human agents (whose reliability rests on shared epistemic norms) (Duede, 2022). No amount of human expertise can make the system transparent (so far).

Essential opacity changes the epistemic practices in a niche. The warranting practices that epistemic niches depend on, what Knorr Cetina (1999) calls “machineries of warranting” (peer review, replication, evidence chains, assessment protocols), assume that the process behind a claim is in principle traceable. A reviewer can ask an author to show their reasoning; a student can be asked to explain how they arrived at an answer; a scientist can be pressed to make their inferential chain explicit. These practices presuppose that knowledge-bearing artifacts in the niche are representationally accessible, even when access requires effort. When essentially opaque artifacts enter the niche, this presupposition breaks down.

AI-generated outputs are plausible but untraceable, shifting the basis for confidence from process to plausibility. Where trust in a claim traditionally rests on being able to trace how it was produced and hold its source accountable, confidence in AI output rests on whether the output looks right (Baier, 1986; Hähnel & Hauswald, 2025). This shift is hard to detect because AI outputs mimic the discursive form of trustworthy sources (well-structured prose, confident tone, citation-like gestures), making plausibility-based confidence feel like process-based trust (Freiman et al., 2025). As AI-generated artifacts accumulate within a niche, warranting practices face pressure from both directions. They assume traceability that essential opacity removes, and AI output’s surface features obscure the fact that traceability is gone. The niche may appear the same, but its warranting infrastructure has been quietly undermined.

3.2 Generativity

Opacity changes how knowledge can be inspected in the epistemic niche. Generativity changes how agents produce within the niche. A notebook stores what you write; a calculator computes what you ask; a search engine retrieves what others have authored. Each of these tools amplifies or extends a cognitive capacity, but the epistemic content originates with human agents. Generative AI produces novel text, code, images, videos, and interpretive analyses. The output is neither fully intended by the human nor the product of another autonomous mind—it typically emerges from the interaction between human prompting and the model’s learned statistical patterns, a form of cognitive production that is neither autonomous nor fully controlled (Barandiaran & Pérez-Verdugo, 2025).

Prior machine-learned models operated within narrow, well-defined parameters. A recommendation algorithm selects from a fixed corpus; a grammar checker applies known rules to existing text; a statistical package computes specified functions on provided data. The human agent could anticipate the space of possible outputs and evaluate them against understood criteria. Generative AI operates across open-ended domains, producing outputs whose content, structure, and framing are not bounded by any fixed rule set. A researcher who asks an LLM to “draft an argument for why this intervention improved outcomes” receives a novel composition that synthesizes and recombines across the model’s training distribution. The space of possible outputs is, for practical purposes, unbounded, and the human agent cannot evaluate the output by checking it against the process that produced it (because of the opacity described above). The narrow parameters that once made machine-generated outputs epistemically manageable are gone.

Besides the unbounded output space, generative AI takes over the open-ended epistemic work through which agents build their understanding of a domain, from drafting arguments to proposing ideas to generating hypotheses. In mathematics, AI systems can now generate candidate proofs, propose conjectures, and cycle through thousands of potential relationships against a dataset—work that once required years of human effort. As Terence Tao observes, AI has driven the cost of idea generation in mathematical discovery to near zero (Patel, 2026). The bottleneck moves downstream: verification, evaluation, and assessing which ideas move a field forward. When AI takes on generative work, inquiry practices reorganize accordingly, with the human role gravitating from producing epistemic content to evaluating, curating, and steering AI-generated content.

This reallocation could in principle be productive; evaluative judgment is itself a sophisticated epistemic capacity (Bearman & Ajjawi, 2023; Boud et al., 2018), and offloading routine generation might free attention for higher-order efforts. But evaluative capacity grows out of generative practice. Human agents learn to judge arguments by constructing them and learn to evaluate code by writing it (Bastani et al., 2025). When developers in a controlled study delegated coding tasks to AI, they showed some productivity gains but significantly impaired conceptual understanding, code reading, and debugging ability (Shen & Tamkin, 2026). Graduate students report similar tensions, relying on AI for literature search, drafting, and code generation while fearing that the convenience erodes the skills a doctorate is meant to build (Nordling, 2026).

The niche-construction perspective scales this concern beyond individual skill loss. Generative work reshapes the epistemic environment—the problems, standards, and practices—that newcomers, collaborators, and future cohorts encounter. If that generative work migrates to AI, the niche may continue to produce outputs while losing the capacity to develop the human agents who can critically evaluate them.

3.3 Substrate asymmetry

Opacity and generativity describe how AI changes what can be inspected and produced within an epistemic niche. Substrate asymmetry explains why it matters whether epistemic work is performed by a human brain or by a machine. Cognition is substrate-dependent. Real-world information processing depends on energy, and energy depends on material substrates (Thagard, 2022). Brains run on neurons and glucose; computers run on silicon chips and electrons. These material differences shape what kind of cognition each substrate supports, and those differences carry epistemic consequences. The *carbon substrate*—brain and body—and the *silicon substrate*—weights and hardware—support epistemic work differently and do different things to the agents performing it.

With generative AI, carbon and silicon substrates are now deeply intertwined in epistemic work. But they operate under fundamentally different conditions. The carbon substrate is metabolic, embodied, and slow; the silicon substrate is electronic, disembodied, and fast. Human agents prompt, steer, and evaluate AI outputs; AI systems generate, extend, and refine human-initiated work. As this coupling deepens, the differences between substrates become consequential.

Before generative AI became mainstream, algorithmic systems already impose machine-paced demands on human agents. Platform algorithms set tight delivery deadlines calculated without regard for slow restaurants, multiple simultaneous orders, or difficult routing conditions; deliverers who miss these deadlines face financial penalties and lower ratings, even when the delay is outside their control (Z. Chen et al., 2022). Generative AI deepens this coupling in more diverse contexts. Where platform algorithms coordinate human labor from the outside, generative AI enters the open-ended epistemic work that was previously reserved for human agents. Knowledge workers now face algorithmic pace pressure in their core intellectual activity—generating insights faster, producing more, accepting outputs they may lack time to scrutinize.

Pace, however, is a surface symptom. The asymmetry that matters for niche construction runs along two deeper dimensions. The first is *finitude*. The carbon substrate operates under important constraints—limited working memory, fatigue, forgetting, slow processing speed—which function as generative pressures. Limited working memory drives chunking and abstraction (Miller, 1956); cognitive load forces representational compression and schema formation (Sweller, 1988); even forgetting serves a productive function by clearing low-utility information and forcing reconstruction. When a student struggles to hold a complex argument in working memory, that struggle spurs the compression and restructuring that constitutes understanding. The silicon substrate has resource constraints of its own (energy consumption, heat dissipation, context window limits), but these are infrastructural, shaping access to computation rather than the character of the “cognition” itself. When AI performs the epistemic work, the epistemic output is produced without the generative constraints that would have shaped it on the carbon substrate—an asymmetry invisible in the output itself.

The second is *plasticity*. The carbon substrate restructures itself through its own activity. Neural connections strengthen, prune, and reorganize in response to the epistemic work the agent performs. This restructuring is a material change that persists beyond the immediate task. The human who struggled through a proof, drafted an argument, or worked through a dataset is differently capable afterward. The silicon substrate is mostly shaped by training and tooling. Its parameters are fixed at training time; what changes during deployment is the data flowing through the system, not the system itself. Fine-tuning and reinforcement learning from human feedback do modify weights, but as design choices external to the model’s own epistemic activity. In-context learning, retrieval-augmented generation, and persistent agent memory change the data flowing through the substrate, not the substrate itself.

Finitude and plasticity are only two dimensions of a broader material difference. Carbon cognition is metabolic, rhythmic, and embodied, coupled to bodily states, shaped by fatigue and incubation, grounded in sensorimotor experience. Silicon cognition is electronic, arrhythmic, and disembodied, operating at nanosecond timescales without metabolic coupling. Neither set of constraints is a deficiency; they are material differences that shape what kind of epistemic work each substrate sustains and how that work reshapes the agent in the process. Existing epistemic niches were constructed around and for carbon substrates. Their practices—peer review, apprenticeship, slow deliberation, iterative revision—reflect carbon temporality and carbon constraints. When a silicon substrate performs generative epistemic work at machine speed and scale, the niche faces pressure to reorganize around silicon capacities, including its pace, tirelessness, and volume of output. If the silicon substrate sets the terms of epistemic work, the practices that sustain carbon-based knowledge building may erode, even as the niche appears more productive by measured outcomes.

The three features compound each other. Opacity means the niche cannot trace how its epistemic content was produced. Generativity means an increasing share of that content originates with AI. Substrate asymmetry means it was produced under conditions alien to the human agents who must interact with it. Together, they create a new epistemic condition: a niche that fills with plausible, novel content at machine speed, while the carbon-based practices for producing and judging such content face pressure from every direction.

4 Enrichment and Erosion of Epistemic Niches

These features described above do not directly determine how an epistemic niche is shaped. How the niche is constructed around these features determines whether the result is *enrichment*, where AI expands human cognitive possibilities, or *erosion*, where AI displaces the generative work through which human capacity is built. Both can emerge in any given niche; the question is which direction the niche is constructed to favor.

4.1 Enriching epistemic niches with AI

Enrichment occurs when AI modifies the epistemic niche in ways that expand cognitive possibilities for human agents without displacing the generative work through which those agents develop their epistemic capacities. AI changes what the niche affords, and the changed niche prompts humans to think and act in ways they otherwise would not have. Different enrichment patterns, operating at different levels of the niche, are visible in emerging practice.

Expanding the epistemic possibility space. AI can surface possibilities that humans have overlooked or dismissed, opening territory for human exploration. Shin et al. (2023) analyzed over 5.8 million moves by professional Go players and found that after the advent of AlphaGo, human players showed increased novelty and quality in their own strategies. The AI demonstrated that moves long dismissed as unviable were in fact strong, and human players then explored those previously neglected regions of the game space, producing innovations of their own. The enrichment lay not in what the AI produced but in what it opened up for human investigation.

This pattern extends beyond games. A similar complementarity is taking shape in mathematical discovery. As Tao describes in an interview (Patel, 2026), AI tools can systematically explore broad problem landscapes, clearing routine observations and identifying areas of genuine difficulty that human experts then investigate in depth. In science, an AI “co-scientist” system generates novel research hypotheses and experimental plans by synthesizing across scientific literatures at a scale no individual

researcher can manage (Gottweis et al., 2025). The system expands the hypothesis space the scientist can consider; the scientist retains the work of judging which hypotheses are worth pursuing, designing experiments, and interpreting results. In medical research, the system proposed novel drug combinations for acute myeloid leukemia that were subsequently validated in vitro (Gottweis et al., 2025). The researcher still does the critical work of evaluating which hypotheses are worth pursuing, designing experiments to test them, and interpreting what the results mean.

Across these cases, the silicon substrate provides breadth that no individual can match; the carbon substrate contributes the cumulative, adaptive reasoning that builds understanding over time. The enrichment works because each substrate contributes what the other cannot.

Restructuring representational infrastructure. Every epistemic niche rests on representational infrastructure—the notational systems, classificatory schemes, and encoding conventions through which people organize and navigate knowledge. Representational innovations like positional notation restructure what their users can think by changing the medium in which cognition operates. AI is producing a comparable restructuring in domains where it introduces fundamentally new representational media.

Architecture illustrates the dynamic. Traditional architectural reasoning operates through Euclidean geometry and stable encoding conventions such as plans, elevations, and axonometric projections. Generative AI reorganizes this representational foundation around “connectionist rationality,” reasoning grounded in probability and statistical inference rather than deduction of form from geometric rules (Huang, 2025). Design similarity is measured statistically rather than geometrically; semantic proximity in a learned feature space replaces geometric distance in Euclidean space. Design becomes an exploratory process of interpolating across latent feature fields rather than composing from geometric primitives.

The restructuring runs deeper than a new tool for an old task. Architectural typologies—the classificatory systems through which architects organize buildings by function, form, and historical lineage—cease to be fixed categories and become probabilistic clusters in latent space. Hybridization and typological drift, previously exceptions requiring theoretical justification, become intrinsic features of the design process (Huang, 2025). Architects can explore transitional forms between building types that the old representational infrastructure could not express. The enrichment lies in cognitive pathways that the restructured representation opens, modes of exploration that had no analogue in the geometric framework.

This pattern, demonstrated in the architectural case, makes the niche construction logic clear. AI modifies the representational conditions of the niche, and when those conditions change, the cognitive operations they support change with them, just as the shift from Roman to Arabic numerals did not add new numbers but made new mathematical operations tractable. The risk is that people may lose fluency with the representational systems the new infrastructure displaces. A niche that can think only through latent space has gained a powerful new medium but narrowed its repertoire if the geometric reasoning that centuries of practice developed atrophies from disuse.

Both enrichment patterns depend on human agents retaining the generative and evaluative work that builds their epistemic capacity. Tankelevitch et al. (2024) argue that generative AI reduces cognitive demands while increasing metacognitive demands—users must become skilled at evaluating and steering AI-generated outputs rather than producing them from scratch. This metacognitive capacity is itself a form of expertise that the niche must cultivate. When users defer to AI-generated proposals rather than exercising independent judgment over them, the enrichment collapses into erosion. The collapse is not binary but gradual. Research on how knowledge workers position themselves relative to AI identifies a spectrum of epistemic relationships, running from instrumental reliance, where AI serves as an efficiency tool with no epistemic authority, through contingent delegation under verified conditions, to authority displacement, where AI is granted partial control over the validation of knowledge (Yang & Ma, 2026). Enrichment holds when users maintain instrumental reliance or contingent delegation, where they retain evaluative authority over AI outputs. It erodes when the relationship drifts toward authority displacement, where AI’s outputs are accepted without independent judgment.

4.2 Eroding epistemic niches with AI

That drift toward displacement is not hypothetical. When AI takes over generative epistemic work without human involvement, the same features that enable enrichment produce erosive feedback loops. The erosion operates at different scales of epistemic niches—individual capacity, community norms, population-level epistemic diversity—and compounds across them.

Capacity erosion through displaced generative work. When AI takes over the generative epistemic work through which human agents build domain understanding, the carbon substrate loses the productive pressures that drive its restructuring. The silicon substrate produces fluent, structured text without the finitude constraints and neural plasticity through which understanding is built. Bastani et al. (2025) found that students with GPT-4 access improved 48% during assisted practice but performed 17% worse on subsequent unassisted tasks, a capacity loss visible only when AI support was removed. Fan et al. (2025) identified a parallel pattern at the metacognitive level. Learners using ChatGPT showed improved task performance while self-monitoring and deep processing declined, undermining the cognitive habits on which long-term knowledge building depends. Oakley et al. (2025) frame the broader dynamic as a “memory paradox” in which offloading cognitive work to

external tools stunts the brain's own knowledge structures, weakening the foundation for reasoning and judgment that AI cannot replace. Zhang and Xu (2025) identify a reinforcing mechanism. Increased AI usage amplifies students' self-efficacy while deepening their dependence, a "paradox of self-efficacy" in which learners feel more capable while becoming less so.

Not all offloading is erosive, of course. When AI handles work peripheral to the core learning task (retrieving known facts, checking formatting, flagging surface errors), the generative pressures that drive understanding remain intact. The erosion begins when AI takes over intrinsic cognitive work such as planning, generating ideas, constructing arguments, and building explanations. Lodge and Loble (2026) describe the result as a "performance paradox" in which AI-assisted output improves while the learner's underlying capacity does not develop, producing "false mastery" that dissociates correct output from conceptual understanding. The erosion falls unevenly. Learners with strong domain knowledge and metacognitive skill can direct AI toward peripheral tasks, retaining the generative work for themselves, while those without such foundations gravitate toward wholesale delegation, widening existing gaps (Chirayath et al., 2025; Lodge & Loble, 2026). Messeri and Crockett (2024) identify an analogous pattern in scientific research, where AI creates "illusions of understanding" as scientists believe they comprehend more than they do when AI serves as "oracle," "surrogate," or "arbiter," substituting plausibility for epistemic contact.

These dynamics amount to negative niche construction, where an environment degrades the builder's own conditions for cognitive thriving. Epistemic niches are volatile. A classroom that has built up sophisticated practices of evidence-based argumentation can lose those practices within a semester if AI tools make the productive effort unnecessary. The erosive loop reinforces itself as AI performs generative work, the carbon substrate's capacity for that work weakens from disuse, the weakened capacity makes AI assistance feel more necessary, and the niche shifts further toward dependence.

Warranting erosion through normalized opacity. AI's essential opacity (see Section 3) breaks the presupposition behind existing warranting practices, namely that the process behind a claim is in principle traceable. The erosive dynamic begins at the point of production. In traditional practice, production itself performs partial warranting. For example, an author who reasons through evidence and commits to inferential steps subjects the developing claim to internal scrutiny that a reviewer can later retrace. If AI dominates production, this built-in warranting disappears. The knowledge worker receives output whose inferential history is opaque and must evaluate it against their own domain understanding. A researcher who uses AI to survey a literature they do not yet command cannot reliably detect when the synthesis selectively emphasizes certain findings or mischaracterizes a position. The warranting challenge is circular because evaluating AI output well requires the expertise that doing the work oneself would have developed.

The pressure escalates as AI-mediated material circulates beyond its original producer. Drafting with AI costs almost nothing; evaluating what it produces demands the same disciplinary expertise as evaluating any scholarly work. When the volume of material entering the workspace outpaces its verification capacity, the practical response is to admit work that would previously have demanded closer scrutiny, normalizing weaker mechanisms of warranting. The result, as Crockett (2026) puts it, is that communities "produce more while understanding less."

Whether erosion or reconstruction prevails depends on whether communities actively develop new warranting practices—adversarial testing, ensemble comparison, peer review that accounts for AI contributions—or passively absorb AI outputs into existing workflows. Active reconstruction of epistemic norms is itself niche construction, modifying the normative environment in response to a new pressure. But without deliberate design, the default is tilted to erosion, because fluent AI output reduces the perceived need for verification.

Diversity erosion through niche homogenization. The dynamics described above operate within individual and group-level epistemic niches. A third pressure operates at the population level. Doshi and Hauser (2024) found that writers using GPT-4 produced individually better stories that were significantly more similar to each other. The finding points to a structural feature of generative AI. Foundation models are trained on overlapping corpora and share statistical tendencies, so outputs drawn from the same model converge toward common framings, rhetorical patterns, and blind spots. When many knowledge workers, research teams, or learning communities depend on the same few models, epistemic niches across a field begin to resemble each other. The convergence is deepened by an "illusion of exploratory breadth" (Messeri & Crockett, 2024). Researchers may believe they are exploring the full space of testable hypotheses when they are actually constrained to the narrower space that current AI architectures optimize.

The erosion here operates at the population level. Cumulative niche construction depends on variation across epistemic strategies—different communities approaching problems differently, maintaining diverse approaches from which novel combinations can emerge (Sterelny, 2012). Homogenization reduces this variation. Individual outputs may improve; collective imagination narrows. Each community's niche may appear enriched—better representations, more fluent discourse—while the population of niches is impoverished, with fewer independent lines of inquiry and less variation in how problems are framed. Designing against this erosion requires attention to how AI-mediated niches relate to each other across a field, beyond how AI functions within any single niche.⁵

⁵The homogenization dynamic may have a dual character. Ghafouri (2026) argues that the same compression that homogenizes outputs within niches also

In summary, enrichment and erosion are not separate futures but concurrent pressures within every epistemic niche that integrates AI. The same AI feature that expands a scientist's hypothesis space can atrophy the disciplinary judgment needed to navigate it; the same representational medium that opens new cognitive pathways can shrink the repertoire if older modes of reasoning fall into disuse. What determines which pressure prevails is not the technology itself but the niche reconstructed with it—the practices, norms, and developmental structures that govern how human agents engage with AI-generated material. And because epistemic niches are inherited and disseminated, the stakes compound across generations and populations. A classroom that normalizes AI-drafted arguments today shapes the epistemic expectations of students who will build tomorrow's research communities. A consultant who helps an organization adopt AI-driven workflows is reconstructing that organization's epistemic niche, embedding assumptions about what counts as knowledge, how decisions should be warranted, and where human judgment remains necessary. The question, then, is not whether AI will reshape epistemic niches but whether the reshaping will be governed by deliberate design. Answering that question requires moving from diagnosis to principles of construction—asking what it takes to build epistemic environments in which AI serves as a substrate for human cognitive development rather than a substitute for it.

5 Discussion

The epistemic niche construction framework shifts the unit of analysis for understanding AI's role in education and society, from individual tools and tasks to the environmental conditions under which humans develop the capacity to learn, inquire, and produce knowledge. Those conditions feed back to reshape the agents who inhabit them. This section draws out implications for educational change, technology design, and future empirical research.

5.1 Implications for educational change

Educational responses to generative AI have followed a pattern familiar from earlier waves of educational technology, absorbing the new tool into existing institutional structures while leaving the underlying model of learning largely intact (Reich, 2020). The first wave focused on detection and restriction—AI text detectors, usage bans, academic integrity policies. A second wave turned to integration, teaching students to prompt effectively, incorporating AI as a writing assistant, redesigning assessments to accommodate AI use (Chan, 2023). Both waves address AI at the level of tools and tasks, asking how technology supports learning within existing educational structures. However, neither engages a more fundamental question: whether students (and educators) retain the capacity to shape the epistemic environments they inhabit.

The niche construction framework foregrounds this question and advances the discussion from epistemic agency to epistemic niche construction. Epistemic agency—forming beliefs, contributing ideas, evaluating claims—operates *within* an existing environment. Epistemic niche construction shapes *the environment itself* by inventing representations, establishing inquiry practices, and building the normative infrastructure that sustains knowledge work. Most educational discussions of AI focus on preserving or enhancing students' agency within AI-mediated environments. The niche construction framework redirects attention to the epistemic environments students inhabit and urges us to intentionally shape them.

This redirection has practical consequences. A design that treats students as users of AI-shaped environments cultivates agency within a niche someone else constructed. A design that positions students (and teachers) as constructors of their own epistemic environments, with AI as one resource among others, cultivates the capacity to shape the conditions of one's own knowledge work. Pedagogical models that center this constructive capacity—Knowledge Building (Scardamalia & Bereiter, 2006), inquiry learning, constructionist approaches—gain renewed significance in the AI era, because they already organize learning around the generative epistemic work most at risk of displacement by AI. When students in a Knowledge Building community negotiate what counts as adequate evidence, develop shared vocabulary for evaluating ideas, or create representational tools that make their collective thinking visible, they are doing epistemic niche construction. When AI takes over this kind of constructive work, the developmental pressure that builds expertise is relieved. The resources for responding to AI's epistemic challenges may already exist within education's own pedagogical traditions.

The epistemic niche construction framework engages the tension between enrichment and erosion at the level of analysis that several scholars have called for. Giannakos et al. (2024) argue for moving beyond "skeptical optimism" about AI in education toward research on ecosystem-level implications; Gibson et al. (2023) call for learning theories that operate across micro, meso, and macro levels. Epistemic niche construction answers both calls by specifying how modifications at one level—a new AI tool in a classroom—propagate through others as changed inquiry practices, altered developmental pressures, and shifted epistemic norms, and by providing the feedback-loop logic that connects environmental change to cognitive change.

lowers translation costs between them, as AI-standardized representational forms allow ideas to travel more easily across disciplinary boundaries. Whether this enables recombinant innovation or simply accelerates convergence depends on whether communities build practices of active curation and recombination into their epistemic environments.

5.2 Implications for technology design

The enrichment and erosion dynamics suggest how learning technologies should be designed.

Scaffold construction over substitution. Because only the carbon substrate is structurally changed by performing cognitive work, technology design should preserve the generative work through which learners build domain understanding—formulating questions, drafting arguments, constructing explanations—while using AI to enrich the informational environment in which that work happens. AI that generates candidate explanations for students to evaluate displaces the productive effort that builds understanding, whereas AI that helps students articulate and refine their own explanations preserves it.

Support meta-representational capacity. When AI expands the epistemic possibility space, the demands on learners shift from production to evaluation and steering. Tankelevitch et al. (2024) argue that generative AI reduces cognitive demands while increasing metacognitive demands. Learners need tools for representing their own representations—examining how their community’s ideas connect, where inquiry has stalled, what assumptions remain unexamined. AI can generate these meta-representations (knowledge graphs, discourse analytics, structural health reports) without performing the interpretive and evaluative work that the representations are meant to support.

Make AI’s epistemic dispositions visible and contestable. When AI restructures representational infrastructure, it embeds epistemic norms that may not match the community’s own. A latent space that measures architectural similarity statistically rather than geometrically silently redefines what counts as “similar” or “different,” privileging dimensions of resemblance that the community may never have endorsed. An AI that synthesizes a research literature foregrounds certain findings and framings while backgrounding others, channeling inquiry along statistically favored paths in ways that can feel like exploratory breadth while narrowing the actual space of investigation. Design should present AI’s framing as one reading among several and support communities in challenging it. An AI that imposes epistemic norms reshapes the niche on the community’s behalf; one that surfaces those norms for scrutiny lets the community reshape its own niche.

Protect variation across epistemic niches. As discussed in Section 4, communities depending on the same foundation models tend to converge on similar framings and strategies (Doshi & Hauser, 2024), and the impression of exploratory breadth that AI creates may make the convergence harder to detect (Messeri & Crockett, 2024). Because convergence can go unnoticed, countering it becomes a design problem. Technology should support each community in adapting AI to its own representational traditions and inquiry practices. The goal is to preserve the variation across epistemic strategies that makes recombination productive.

5.3 Implications for empirical research

Two research programs follow from this analysis. The first is *cognitive ethnography of AI-integrated epistemic niches*. Following Nersessian’s (2022) methodological model for studying how epistemic practices develop in research laboratories, researchers could document how the representations, practices, and values of learning communities are restructured when AI tools are introduced. Empirical studies are needed to trace what gets modified, how the modification feeds back to reshape agents, and over what timescale. Most existing studies of AI in education measure short-term performance differences. What the framework calls for is situated and longitudinal observation of how epistemic niches develop, stabilize, or degrade over the course of a semester or longer, tracking whether enrichment or erosion trajectories unfold in practice.

The second concerns *the relationship between production and evaluation*. The niche construction framework predicts that offloading generative epistemic work to AI will weaken evaluative capacity over time, because the work of generating ideas, arguments, and explanations is what builds the domain understanding on which evaluation depends. This prediction is testable. Studies could compare the evaluative judgment of learners who have extensive production experience in a domain with those whose production has been substantially AI-assisted, examining whether the capacity to assess idea quality, identify promising lines of inquiry, or detect flawed reasoning differs between the two groups. The Bastani et al. (2025) finding that AI-assisted practice degraded subsequent unassisted performance is suggestive, but the mechanism linking production experience to evaluative capacity remains to be traced in detail.

5.4 Limitations

The analysis as presented has several limitations that warrant acknowledgment. First, the paper has focused on epistemic niche construction as it operates in educational and research contexts without addressing who gets to construct epistemic niches and whose constructive capacity is supported or undermined by current AI deployments. The relationship between epistemic niche construction and epistemic justice (Fricker, 2007) deserves sustained treatment that this paper cannot provide. Foundation models trained predominantly on English-language, Western-sourced corpora embed epistemic norms that may systematically disadvantage communities whose knowledge traditions are underrepresented in training data (Kay et al., 2024), and algorithmic systems in education have already been shown to reproduce and amplify existing biases along lines of race, gender, and socioeconomic status (Baker & Hawn, 2021). Communities differ in their access to AI tools, in the resources available for critical evaluation of AI outputs, and in the power to shape the AI systems that increasingly structure their

epistemic environments. A fuller account would examine how niche construction intersects with existing inequalities in who gets to shape the conditions of learning and knowledge production.

Second, the substrate comparison risks being read as an essentialist claim that carbon cognition is inherently superior to silicon processing. The point is not that human cognition is “better” but that only the carbon substrate is structurally changed by performing cognitive work, with implications for how cognitive labor should be distributed in environments designed to develop human capacity.

Third, the paper has treated “generative AI” as a relatively stable category. In practice, AI systems are developing rapidly, and the features identified in this analysis—opacity, generativity, substrate asymmetry—may shift as the technology changes. Advances in interpretability research could reduce essential opacity; new training paradigms could alter the generativity landscape; developments in neuromorphic computing could narrow the substrate asymmetry. The analysis’s value lies in directing attention to the deeper questions—what is modified, how does the modification feed back, and over what timescale—rather than in any fixed characterization of AI’s current capabilities. The most consequential of those questions remains the one this paper began with: how to support humans in constructing epistemic niches that sustain their capacity to think, learn, and build knowledge, even as AI reshapes the conditions under which that construction proceeds.

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Appendix: Statement on AI Use

In the process of developing this paper, AI tools including Claude Code and Google Gemini were used. The author used Claude as a writing collaborator for drafting prose from outlined ideas, revising for clarity and cohesion, and searching academic literature via Zotero and online databases (using MCP plugins). All intellectual direction—the arguments, structure, and theoretical commitments—originated with the author and long predated the writing itself. Claude executed the author's instructions: drafting text from specified outlines, revising passages according to explicit editorial guidance, retrieving relevant literature from Zotero, and formatting citations. Solitary thinking, based on carbon substrate, was relied on to weave together the vision and argument. Discussions with other scholars (e.g., through a conference talk and follow-up conversations) were instrumental in refining this work.

To illustrate the nature of this collaboration, the author's prompts were extracted from Claude Code session logs. The logs (223 prompts across 7 sessions) include the author's instructions during the intensive writing process. Overall, the median

prompt length was 129 characters; most prompts (51%) fell between 50 and 200 characters. Prompts fell into several categories: editorial revision and critique (59%), brief directives such as confirmations or continuation requests (21%), literature search and citation management (10%), and structural direction on argument organization (8%). The examples below illustrate the range of author involvement. They are organized by type, not chronologically.

Editorial revision and critique:

“line 5 reads a lot like what we want to avoid. i do like the message – we need to go deeper than being more efficient in the old, dated model, to rethink what learning needs to be as AI introduces something fundamentally different for human cognition and society.”

“section 4 now reads well in general. now i want you to do a few more passes, identifying overlaps that could be compressed, enhancing cohesion. 4.2 may have some paragraphs that may need some tightening. and we need a final paragraph to sum up the whole section.”

“ ‘The scientist’s role shifts toward evaluation, refinement, and experimental testing, but the epistemic work remains with the human agent.’ - guess the scientist and the human agent are the same. but the sentence structure is not hinting this.”

Literature research:

“i reviewed section 3. there is an important note at the beginning of the section that needs attention. now please take time to work on it, as well as other minor notes i added. use zotero mcp to locate referenced works when needed.”

Conceptual pushback:

“the asymmetry is not about one is learning/changing and the other is not. it is about one does things differently than the other, with different constraints, temporality, etc. they are just different.”

“for this whole point, i am not sure monitoring is the right way to go. what’s the higher level principle that is more aligned with niche construction?”

A paper arguing that AI reshapes the epistemic conditions under which humans produce knowledge should be transparent about the epistemic conditions under which it was itself produced. This appendix is offered in that spirit. The account above is illustrative rather than exhaustive. Other AI-mediated interactions shaped the writing process in ways that are harder to track. These interactions included Google Search’s AI overviews, casual exchanges with Claude mobile app, focused discussions with local LLMs on an international flight, etc. A full accounting of AI’s contribution to any piece of knowledge work may be inherently incomplete, which is itself a consequence of the epistemic niche restructuring this paper describes.